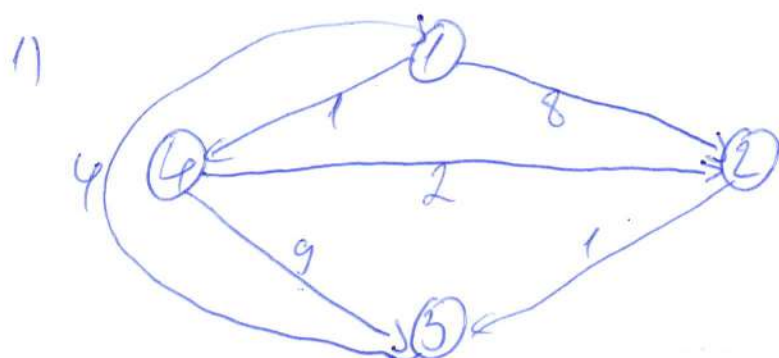


Laborator 11



Matrice costuri

	1	2	3	4
1	0	8	∞	1
2	∞	0	1	∞
3	4	∞	0	∞
4	∞	2	9	0

$$D^0 = \begin{pmatrix} 0 & 8 & \infty & 1 \\ 2 & 0 & 1 & \infty \\ 4 & \infty & 0 & \infty \\ \infty & 2 & 9 & 0 \end{pmatrix}$$

$$\text{for } k = 1, 4$$

$$\text{for } i = 1, 4$$

$$\text{for } j = 1, 4$$

$$\text{if } j < \min(d_{ij}, d_{ik} + d_{kj})$$

$$D^1 = \begin{pmatrix} 0 & 8 & 2 & 1 \\ 2 & 0 & 1 & \infty \\ 4 & 12 & 0 & 5 \\ 2 & 2 & 9 & 0 \end{pmatrix}$$

Linia 1, coloana 1 si diagonala nămăm meschiambote

$$k=1$$

$$i=1$$

$$j=2, j=1, j=2$$

$$j=3 \min(d_{13}, d_{11} + d_{13})$$

$$1 < 2 + \infty$$

$$j=4 \min(d_{14}, d_{11} + d_{14})$$

$$i=3$$

$$j=2 \min(d_{32}, d_{31} + d_{12})$$

$$2 > 4 + 8$$

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$$1) L=4 \min(d_{n4}, d_{n1} + d_{14})$$

$$8 > 4 + 1$$

$$i=4$$

$$L=2 \min(d_{n2}, d_{n1} + d_{12})$$

$$2 < 8 + 8$$

$$L=3 \min(d_{n3}, d_{n1} + d_{13})$$

$$9 < 8 + 4$$

$$D^2 = \begin{pmatrix} 0 & 8 & 9 & 1 \\ 8 & 0 & 1 & 8 \\ 4 & 12 & 0 & 5 \\ 8 & 2 & 3 & 0 \end{pmatrix}$$

$$k=2, i=1, L=3 \min(d_{n3}, d_{n2} + d_{23})$$

$$2 > 8 + 1$$

$$L=4 \min(d_{n4}, d_{n2} + d_{24})$$

$$1 < 8 + 8$$

$$i=2$$

$$i=3$$

$$L=1 \min(d_{n1}, d_{n2} + d_{21})$$

$$4 < 12 + 8$$

$$L=4 \min(d_{n4}, d_{n2} + d_{24})$$

$$5 < 12 + 8$$

$$i=4$$

$$L=1 \min(d_{n1}, d_{n4} + d_{41})$$

$$8 = 2 + 8$$

$$L=3 \min(d_{n3}, d_{n4} + d_{43})$$

$$9 > 2 + 1$$

$$D^3 = \begin{pmatrix} 0 & 8 & 9 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 12 & 0 & 5 \\ 7 & 2 & 3 & 0 \end{pmatrix}$$

$$k=3, i=1$$

$$L=2 \min(d_{n2}, d_{n3} + d_{32})$$

$$8 < 9 + 12$$

$$L=4 \min(d_{n4}, d_{n3} + d_{34})$$

$$1 < 9 + 5$$

$$i=2$$

$$1) L = 1 \min(d_{21}, d_{23} + d_{31})$$

$$2 > 1 + 4$$

$$L = 4 \min(d_{24}, d_{23} + d_{34})$$

$$5 > 1 + 5$$

$$i = 4$$

$$L = 1 \min(d_{41}, d_{43} + d_{31})$$

$$2 > 5 + 4$$

$$L = 2 \min(d_{42}, d_{43} + d_{32})$$

$$2 < 5 + 0$$

$$D^4 = \begin{pmatrix} 0 & 3 & 4 & 1 \\ 5 & 0 & 16 \\ 4 & 7 & 0 & 5 \\ 7 & 2 & 3 & 0 \end{pmatrix}$$

$$k = 4$$

$$i = 1$$

$$L = 2 \min(d_{17}, d_{14} + d_{42})$$

$$2 > 1 + 2$$

$$L = 3 \min(d_{13}, d_{14} + d_{43})$$

$$9 > 1 + 3$$

$$i = 2, L = 1 \min(d_{24}, d_{23} + d_{31})$$

$$5 < 1 + 7$$

$$L = 3 \min(d_{23}, d_{24} + d_{43})$$

$$1 < 6 + 0$$

$$i = 3, L = 1 \min(d_{31}, d_{34} + d_{41})$$

$$4 < 0 + 7$$

$$L = 2 \min(d_{32}, d_{34} + d_{42})$$

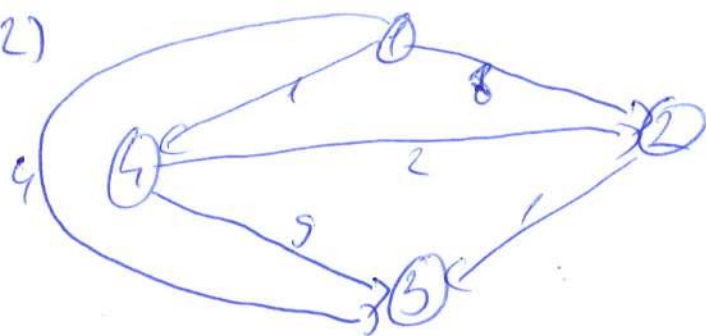
$$12 > 5 + 2$$

Drumurile de cost minim

	1	2	3	4
1	0	3	4	1
2	5	0	16	
3	4	7	0	5
4	7	2	3	0

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2)



$$\Delta^0 = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Let $k = 1, 4$

Let $i = 1, 4$

if $(d_{ik} = 0)$ then

Let $L = 1, 4$

$d_{ik} \leftarrow d_{ik} \vee d_{kL}$

$$\Delta^1 = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

$k=1$

$i=1 \quad d_{11} = 1A$

$L=2 \quad d_{12} = d_{12} \vee d_{12} = d_{12} = 1$

$L=3 \quad d_{13} = d_{13} \vee d_{13} = d_{13} = 0$

$L=4 \quad d_{14} = d_{14} \vee d_{14} = d_{14} = 1$

$i=2 \quad d_{24}$ pe linia 2 se copiază linia 2 din Δ^0

$i=3 \quad d_{31} \leftarrow 1A$

$L=1 \quad d_{31} \leftarrow d_{31} \vee d_{31} = 1$

$L=2 \quad d_{32} \leftarrow d_{32} \vee d_{32}$

$0 \vee 1 = 1$

$L=4 \quad d_{34} \leftarrow d_{34} \vee d_{34}$

$0 \vee 1 = 1$

$i=4 \quad d_{41} = 1F$ se copiază linia 4 din Δ^0 in Δ^1

$$\Delta^2 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

2) $K = 2$

$i = 1 \quad d_{12} = 1A$

$L = 2 \quad d_{12} \in d_{12} \vee d_{22}$
 $1 \vee 1 = 1$

$L = 3 \quad d_{13} \in d_{13} \vee d_{23}$
 $0 \vee 1 = 1$

$L = 4 \quad d_{14} \in d_{14} \vee d_{24}$
 $1 \vee 0 = 1$

$i = 2 \quad d_{22} = 1A$

$L = 1 \quad d_{21} \in d_{21} \vee d_{22}$
 $0 \vee 1 = 1$

$L = 4 \quad d_{24} \in d_{24} \vee d_{22}$
 $0 \vee 1 = 1$

$i = 3 \quad d_{32} = 1A$

$L = 1 \quad d_{31} \in d_{32} \vee d_{21}$
 $1 \vee 1 = 1$

$L = 4 \quad d_{34} \in d_{34} \vee d_{24}$
 $1 \vee 0 = 1$

$i = 4 \quad d_{42} = 1A$

$L = 1 \quad d_{41} \in d_{42} \vee d_{21}$
 $0 \vee 1 = 1$

$L = 2 \quad d_{42} \in d_{42} \vee d_{22}$
 $1 \vee 1 = 1$

$L = 3 \quad d_{43} \in d_{43} \vee d_{23}$
 $1 \vee 1 = 1$

$S^3 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$

Verificam doar pozitiile unde valoarea este 0

$K = 3$

$i = 2 \quad d_{23} = 1A$

$L = 1 \quad d_{21} \in d_{21} \vee d_{23}$
 $0 \vee 1 = 1$

$L = 4 \quad d_{24} \in d_{24} \vee d_{23}$
 $0 \vee 1 = 1$

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$$2) i=4 \text{ dus } = 1A$$

$$A = 1 \text{ d } 41 \in \text{d } 41 \text{ v } d \text{ n } 1$$

$$0 \text{ v } 1 = 1$$

Toate matricele D^3 este plina de val 1 $\Rightarrow$

$$\Rightarrow \Delta^4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$